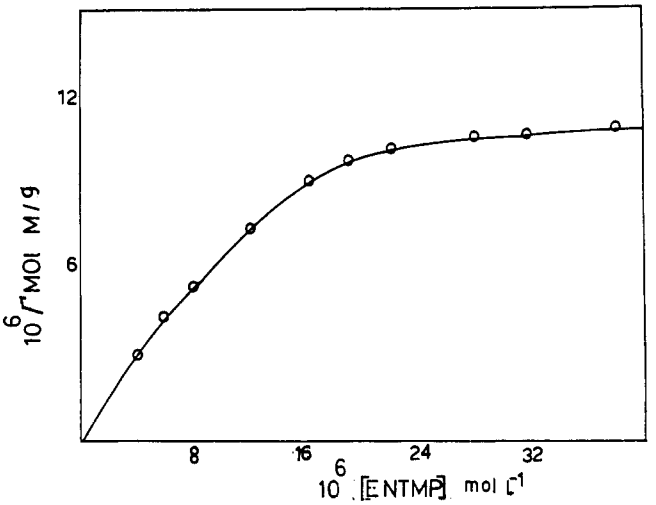


Figure 6. Adsorption of ENTMP on calcium fluoride crystals at equilibrium. Plots of Γ against [ENTMP].

uration σ = 0.4, 0.2, and 0.1 respectively. These values reflect the high adsorption affinity at low relative undersaturation in the presence of ENTMP. A similar dependence of the degree of inhibition with change in driving force has been observed for the influence of phosphonate on the dissolution rate of magnesium fluoride[17] and barium fluoride[19] in aqueous solution. As noted for the crystallization of gypsum by Van Rosmalen and coworkers,[20] the effectiveness of the HEDP as an inhibitor depends on the degree of supersaturation.

In order to investigate the adsorption of phosphonate on the calcium fluoride crystals surface, adsorption equilibrium experiments were made for ENTMP on calcium fluoride crystals at σ = 0. A typical adsorption isotherm is plotted in Figure 6. Assuming that the area occupied by an ENTMP molecule is 50 × 10^{-20} m^2, the fraction of the solid surface covered by adsorption molecules is only 6% at the plateau in Figure 6. At this ENTMP concentration, the dissolution rate is reduced by more than 92%. Adsorption affinity constant at equilibrium 5.2 × 10^5 dm^3 mol^{-1} is in satisfactory agreement with the kinetic adsorption affinity constants 10^5 dm^3 mol^{-1}. The dissolution rate of magnesium fluoride was reduced by more than 80% when only 9% of the crystal surface was covered by HEDP molecules.[17] Moreover, the rate of crystallization may be reduced virtually to zero when only 5–7% of the crystal surface was covered by adsorbed inhibitor molecules.[21,22]

Registry No. HEDP, 2809-21-4; ENTMP, 1429-50-1; CaF_2, 7789-75-5; NaCl, 7647-14-5; KH_2PO_4, 7778-77-0.

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Ion Recombination Rates in Rare-Gas Cation–Halide Anion Systems. 2. KrF* and XeCl*

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The emission spectroscopy/pulse radiolysis method of determining three-body ionic recombination rate constants in rare-gas–halogen source gas mixtures has been extended to systems where the emission is produced by both ionic and nonionic pathways. This has enabled recombination coefficient measurements to be done over a large pressure range for irradiated Kr/SF_6 and $Xe/CFCl_3$ gas mixtures. The rate constants measured for both these systems show the typical pressure dependence of an increase to a maximum value of ~2.5 × 10^{15} M^{-1} s^{-1} (~4 × 10^{-6} cm^3 s^{-1}), before the onset of the diffusion-controlled reaction. These values have been compared to the predictions of the Langevin–Harper diffusion-controlled and the Bates termolecular recombination models. The large discrepancies between theory and experiment have shown that other recombination processes dominate the ionic recombination.

Introduction

The electron pulse radiolysis work[1-7] done on the rare-gas–halogen source gas (R/AX) mixtures has shown that under suitable experimental conditions, the general mechanism for exciplex formation is

$$R \rightsquigarrow R^*, R^+, e^-_{(s)} \quad \text{initiation} \quad (1)$$

$$e^-_{(s)} + M \rightarrow e^-_{(th)} \quad \text{hot electron thermalization (M = R or AX)} \quad (2)$$

$$e^-_{(th)} + AX \rightarrow AX^-/X^- \quad \text{thermal electron capture} \quad (3)$$

$$R^* + AX \rightarrow RX^* + A \quad \text{direct reaction of rare-gas excited states} \quad (4)$$

$$R^+ + 2R \rightarrow R_2^+ + R \quad \text{cation dimerization} \quad (5)$$

$$R_2^+ + AX^-/X^- (+M) \rightarrow RX^* + \text{products} \quad \text{three-body ionic recombination} \quad (6)$$

$$RX^* \rightarrow R + X + h\nu \quad \text{excimer fluorescence} \quad (7)$$

In a previous investigation[8] the techniques of pulse radiolysis and emission spectroscopy were used to determine the reaction 6 rate constants for irradiated Xe/SF_6 gas mixtures. This system was chosen for initial study as the XeF* exciplex was formed *only*

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by ionic recombination. Rate constants were obtained over a very large pressure range and contrasted with the predictions of current recombination theories. Excellent agreement with the Langevin–Harper[9,10] theory was found in the diffusion-controlled limit, but at lower gas pressures the Bates[11] model predictions were much lower than the data experimentally observed.

In this work an extension of this method for the determination of ion–ion recombination rate constants in the formation of the KrF* and XeCl* exciplexes is reported. For these two species, account has to be taken of the significant proportion of the exciplex formed by reaction 4. The kinetics of this process can overlap some of the earlier stages of the ionic recombination process. The successful resolution of these two processes was achieved, and the derived rate constants were compared to current theoretical predictions.

Experimental Section

The established pulse radiolysis facilities[8,12] in the Department of Chemistry at the University of Melbourne were used for this study. The conditions of this study were identical to those stated previously,[8] and thus only a brief description will be given here.

The irradiation device used for all this work was a Field Emission Corp. Febetron 706 electron pulser. The exciplex emissions from the irradiation cell were monitored perpendicular to the electron beam. A beam splitter and two monochromator/photomultiplier tube setups were used to simultaneously generate kinetic and time-integrated traces on two calibrated Tektronix 7633 storage oscilloscopes. The kinetic traces were recorded with 50-Ω termination, while the integrated signals were generated by terminating the photomultiplier output into the input impedence of the oscilloscope, (~1 MΩ and 20 pF). A sample and hold circuit was placed in parallel with the integrating oscilloscope to measure the total amount of light emitted.

All oscilloscope traces were digitized and stored via a TV camera interfaced to a computer. The resulting data were stored on a dual floppy disk system and later analyzed.

All the gas manipulations were done in a conventional mercury free metal/glass vacuum line. Gas cylinders and the irradiation cells were connected to the vacuum line via stainless steel or brass Cajon Ultra-Torr fittings. Purified gas samples were contained in storage bulbs attached to the line itself. The filling of irradiation cells was done by direct expansion of the gas into the cell. All pressures were measured on a MKS Baratron pressure gauge, type 170M-6B (0.05–1200 Torr) or by a Budenburg AS1349 pressure gauge (0–600 kPa).

The gases used in this study were krypton (Matheson research grade (99.995%)), xenon (Matheson research grade (99.995%)), SF_6 (Matheson pure grade (99.9%)), $CFCl_3$ (C.I.G. pure grade), and CF_2Cl_2 (C.I.G. pure grade) and were subjected to several freeze (77 K)–pump–thaw cycles prior to usage.

Experiments were conducted to see whether repetitive pulsing of the gas sample caused significant degradation. It was found in all gas mixtures that although the total amount of light being emitted at any wavelength remained the same for up to 20 pulses, the kinetics of emission changed significantly (>10%) from the first pulse. Hence all kinetic data were recorded from the first pulse.

All irradiations were done at room temperature.

Data Analysis. For ionic recombination as the *only* exciplex formation process, it has previously been shown that[7]

$$\frac{1}{I^{1/2}} = \frac{1}{K^{1/2}}\left(\frac{1}{\alpha^{1/2}[R^+]_0} + \alpha^{1/2}t\right)$$

where I is the photon intensity observed by the detection system, K is a proportionality constant for the light detection efficiency of the experimental setup, and α is the recombination rate constant. Thus a plot of $1/I^{1/2}$ vs t should be a straight line of slope $m = \alpha^{1/2}/K^{1/2}$.

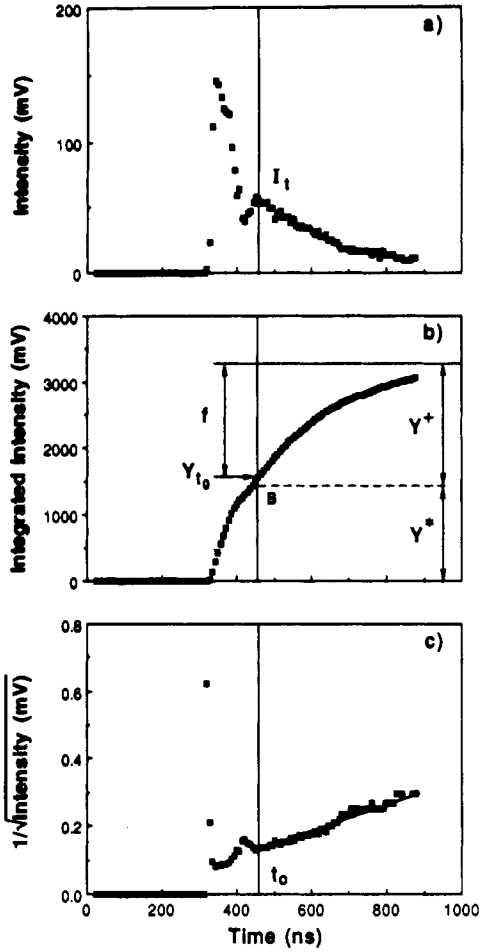


Figure 1. (a) Typical emission curve at 248 nm observed from an irradiated Kr (40.0 Torr)/SF_6 (0.50 Torr) gas mixture. I_t: the intensity of the kinetic trace (mV) at time t_0. (b) Resolution of the relative yields of the ionic recombination (Y^+) and direct reaction fluorescence (Y^*) from the discontinuity (B) in the integrated measurement of the kinetic trace of (a). (c) Transformed kinetic emission curve with straight-line analysis beginning at time t_0.

The photon intensity, I_t, at any time, t, is given by

$$I_t = K\alpha[Kr_2^+]_t[SF_6^-]_t$$

and by charge balance

$$[Kr_2^+]_t = [SF_6^-]_t = n_t$$

Substituting for $K^{1/2}$ and rearranging gives

$$\alpha = m(I_t^{1/2}/n_t)$$

Thus to obtain α, the gradient of the transformed kinetic plot, m, must be multiplied by the square root of the observed emission intensity (mV) at any time t, I_t, and divided by the ion concentration, n_t, at this time.

Figure 1a shows the emission–time curve from 40.0 Torr of krypton and 0.50 Torr of SF_6 at 248 nm. To obtain the gradient m, the emission intensity was transformed as described above (see Figure 1c). As expected, the presence of the extraneous emissions caused the initial part of the transformed curve to be nonlinear.

The analysis of this transformed curve, by linear least-squares fitting, was done only on the limiting linear portion, and the time, t_0, at which the analysis began, noted. From time t_0 onward there was *only* ion recombination formed fluorescence occurring. This was verified from separate observations of the dimer rare-gas fluorescence lifetime under the same rare-gas pressure conditions and also by calculation of the time taken to complete production of the exciplex by the rare-gas neutral excited-state reaction. It

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has been previously shown[12] that the excited state formed fluorescence can be accurately modeled by the following reaction scheme:

$$Kr^* + SF_6 \rightarrow KrF^* \qquad k_a = 1.2 \times 10^{12}\ M^{-1}\ s^{-1}\ [13] \qquad (a)$$

$$Kr^* + Kr \rightarrow \text{products} \qquad k_b = 1.0 \times 10^{10}\ M^{-1}\ s^{-1}\ [13] \qquad (b)$$

$$Kr^* \rightarrow h\nu \qquad k_c = 5.05 \times 10^{6}\ s^{-1}\ [13] \qquad (c)$$

$$KrF^* + SF_6 \rightarrow \text{products} \qquad k_d = 3.6 \times 10^{12}\ M^{-1}\ s^{-1}\ [12] \qquad (d)$$

$$KrF^* + Kr \rightarrow \text{products} \qquad k_e = 8.6 \times 10^{9}\ M^{-1}\ s^{-1}\ [14,15] \qquad (e)$$

$$KrF^* + 2Kr \rightarrow \text{products} \qquad k_f = 2.1 \times 10^{11}\ M^{-2}\ s^{-1}\ [15-17] \qquad (f)$$

$$KrF^* \rightarrow h\nu \qquad k_g = 1.48 \times 10^{8}\ s^{-1}\ [15] \qquad (g)$$

Rate constants k_e and k_f are averaged over the values reported in these papers.

For example, under the conditions of Figure 1a (40.0 Torr of Kr, 0.50 Torr of SF_6), the time needed for excited state produced KrF* fluorescence to become negligibly small (<1%) is ~100 ns. The value of t_0, as shown for this example, was typically chosen at a later time than the minimum value calculated from the reaction scheme.

The value of I_t was the intensity of the emission–time trace at time t_0, as seen in Figure 1a.

To determine the ion concentration at t_0 the following procedure was used. The emission–time trace was simultaneously integrated, as described before. This trace was found to be composed of several components: X-rays, dimer rare-gas fluorescence, exciplex fluorescence formed by the reaction of rare-gas excited states, and exciplex fluorescence formed by ionic recombination.

The temporal dependence of the emission on initial pulse intensity, added SF_6 pressure and total pressure gave the following observations:

The X-ray signal followed the time profile of the electron pulse (~5 ns) and was typically only a few percent of the signal.

The dimer rare gas fluorescence, from Kr_2^*, was typically complete within 100 ns, but its intensity varied greatly with rare-gas pressure. At very low pressures (<100 Torr), its intensity was low, ~10% of the signal, but at higher pressures (>1000 Torr), it accounted for more than 90%. This background fluorescence determined the upper pressure limit of the measurements done in this study; when the fluorescence due to background Kr_2^* was greater than 90% of the total, its subtraction from the total emission gave large errors and poor reproducibility. This upper limit was reached at a krypton pressure of ~1600 Torr and a xenon pressure of ~2400 Torr.

The exciplex fluorescence formed by the reaction of electronically excited states of krypton with SF_6 was again complete typically within 100 ns. However, it was more intense at low rare-gas pressures, where it could account for up to 70% of the integrated signal intensity.

The ionic recombination formed exciplex fluorescence had the slowest *rate* of production, being observed for many hundreds of nanoseconds. Its intensity was also dependent on total gas pressure, being comparable to the direct reaction fluorescence intensity at low gas pressures but accounting for nearly 100% of the exciplex fluorescence at high gas pressures.

From this integrated signal, the X-ray and dimer rare-gas fluorescence intensities were subtracted. This was done exactly as for the Xe_2^+/SF_6^- recombination system,[8] by irradiating the gas mixture a second time to measure the total intensity at a wavelength just outside the exciplex emission band. A separate irradiation of only the rare gas correlated the emission intensity at this wavelength to the intensity at the wavelength at which the kinetic measurement was made.

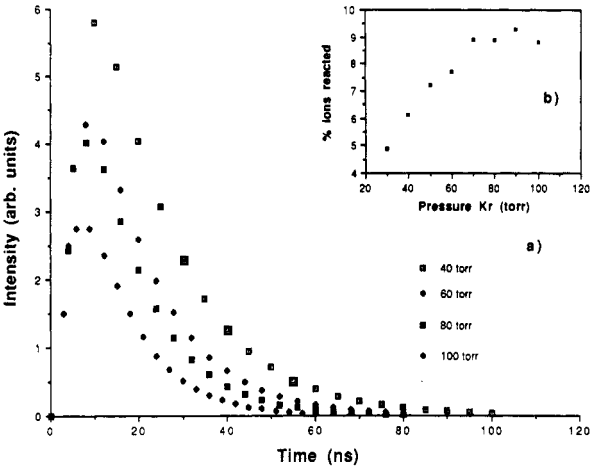


Figure 2. (a) Krypton pressure dependence on the temporal behavior of the fluorescence produced by reaction of krypton electronically excited states with SF_6. Simulation conditions: $[SF_6] = 2.71 \times 10^{-5}$ M (0.50 Torr), $[Kr^*]_{t=0} = 1.0 \times 10^{-9}$ M. (b) Pressure dependence of the percentage of ionic recombination that has occurred during the time taken for the excited-state produced fluorescence to become negligibly small (<1%).

After subtraction, the remaining emission was as shown in Figure 1b, consisting of krypton excited neutral and ionic recombination formed exciplex fluorescence.

The slight discontinuity, B, seen in this integral curve (Figure 1b) indicates the end of the production of photons by the excited-state mechanism. From this the relative yields of the excited-state fluorescence, Y^*, and ionic recombination fluorescence, Y^+, can be determined. Since the initial ion concentration is known from ozone dosimetry[18] and stopping powers[19] of the constituent gases, the remaining ion concentration at any time, t_0, *after* this discontinuity, can thus be determined from the fraction f (Figure 1b):

$$f = (Y_\infty - Y_{t_0})/Y^+$$

where Y_∞ is the sample and hold measurement, and Y_{t_0} is the integrated emission intensity at time t_0.

It may sensibly be asked at this point what error is involved in assuming that the discontinuity is the point where all the excited-state fluorescence has ceased and that *no* ionic recombination fluorescence has yet occurred.

The relative yields of neutral excited-state and ionic recombination fluorescence could only be observed up to ~100 Torr of krypton. Over this pressure range the time dependence of the neutral exciplex fluorescence, as modeled by the above reaction scheme, is shown in Figure 2a as a function of krypton pressure. With the ionic recombination rate constants calculated in this study by the above procedure (from Figure 4a), the amounts of ionic recombination fluorescence over this time scale is small (see Figure 2b).

The experimentally observed discontinuity is seen to occur approximately 105 ns after the start of the trace, in excellent agreement with the calculated end of the exciplex fluorescence produced by neutral reaction. This confirms the use of this method as a measure of the yield produced by excited states of krypton. Even if this estimate of the excited state yield is in error by 30%, the calculated recombination rate constant would only be in error by ~14%. Furthermore, this error has its maximum value at these low pressures and quickly becomes far smaller than the experimental scatter as the pressure is increased.

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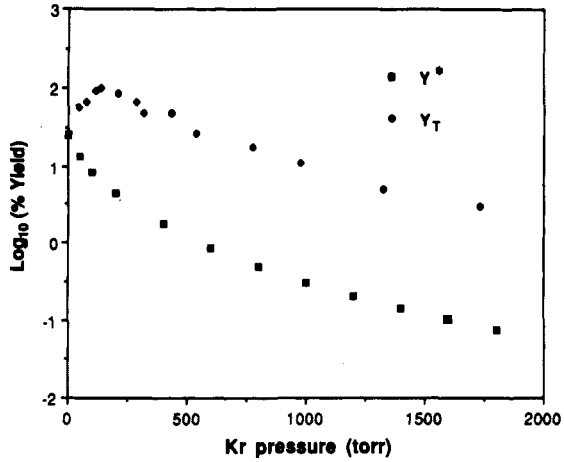


Figure 3. Krypton pressure dependence of the measured total (Y_T) and calculated excited-state reaction (Y^*) yields for Kr_2^+/SF_6^- recombination.

The relative yields measured under these conditions were used together with the mechanism given above to calculate the relative yield of photons from the neutral reaction process over the entire krypton pressure range studied. It is found that at low pressures, where the excited-state fluorescence lives the longest and the Y^*/Y^+ ratio is the largest, the recombination rate constants are low and have first $\tau_{1/2}$ values in the microsecond range. At higher pressures, where the α values are much larger, the excited-state reaction fluorescence is very short lived due to quenching and is of low yield. This is shown in Figure 3, where the experimental yields of the total fluorescence are contrasted to the calculated yields of the excited-state reaction fluorescence.

In summary, the resolution of excited-state and ionic recombination formed exciplex fluorescence yields by this procedure is valid and introduced only small error in the calculation of ionic recombination rate constants.

Results and Discussion

(1) Low/Medium Pressures. (a) Kr_2^+/SF_6^- Ionic Recombination. The emission from the $^2\Sigma^+_{1/2} \rightarrow {}^2\Sigma^+_{1/2}$ transition in the KrF* molecule[20] covers the wavelengths 234–254 nm with an emission peak at 248 nm. All kinetic measurements done for this system were at the peak wavelength.

The experimental conditions ensured the following:

(1) Electron thermalization (by both gases) was complete within the duration of the electron beam pulse (~5 ns). The thermalized electrons, $e^-_{(th)}$, were then rapidly captured by SF_6 (0.50 Torr, $k = (1.37 \pm 0.04) \times 10^{14}$ M^{-1} s^{-1},[21] (2.28×10^{-7} cm^3 s^{-1})) to form the molecular anion, SF_6^-, which was stable over the time domain of these experiments[22] (0.5–2.0 μs).

(2) The initial ion concentration was kept very low (10^{-9}–10^{-8} M, (5–50) $\times 10^{11}$ cm^{-3}) and fast ionic dimerization of the rare-gas cation was ensured by using high enough krypton pressures and/or low initial pulse intensity and long time scales. At most of the pressures used in this study (400 Torr or greater), the half-life for dimerization was far shorter than that for ion recombination. For the pseudo-first-order reaction

$$Kr^+ + 2Kr \rightarrow Kr_2^+ + Kr \qquad k = 8.70 \times 10^{10}\ M^{-2}\ s^{-1}$$ [23]

At 400 Torr of krypton

$$k = 4.07 \times 10^7\ s^{-1}$$

i.e.

$$\tau_{1/2} \approx 17\ ns$$

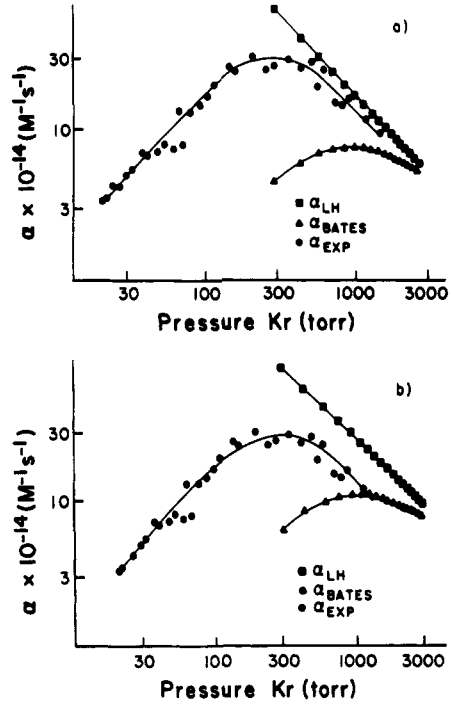


Figure 4. (a) Comparison of the experimental rate constants, α_{EXP}, to the termolecular Bates, α_{BATES}, and diffusion-controlled Langevin–Harper, α_{LH}, calculations for the reaction $Kr_2^+ + SF_6^- + Kr \xrightarrow{\alpha}$ products. (b) As for (a) but for the reaction $Kr_2^+ + F^- + Kr \xrightarrow{\alpha}$ products.

For ion recombination reaction under these conditions the first half-life is given by

$$\alpha = 2.80 \times 10^{15}\ M^{-1}\ s^{-1}$$

where typically

$$n_0 \sim 5 \times 10^{-9}\ M$$

i.e.

$$\tau_{1/2} \approx 71\ ns$$

At lower krypton pressures, complete cation dimerization was ensured by observing the recombination on the microsecond time scale. The removal of the dimer, and/or the monomer cation by charge transfer to SF_6 was impossible, as the ionization potential of SF_6 (15.4 eV[24]) is higher than that of krypton (13.999 eV[25]).

The correction of the integrated intensity for krypton dimer fluorescence and the signal from X-rays was made as described, using a background wavelength of 255 nm.

The recombination coefficients, as a function of krypton pressure, are shown in Figure 4a. These values show the characteristic pressure profile observed in previous ion–ion recombination rate constant determinations over this pressure range,[8,26,27] with a continuous increase up to a maximum value (2.9×10^{15} M^{-1} s^{-1}, ~4.8×10^{-6} cm^3 s^{-1}) at approximately 300 Torr, followed by a decrease at higher krypton pressure.

By analogy to these previous works, the ionic recombination mechanism at low pressures was believed to be two-body, i.e.

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$$Kr_2^+ + SF_6^- \rightarrow KrF^* + \text{other products}$$

and the increase in rate constant at medium pressures is attributed to the increasing contribution of the three-body recombination mechanism

$$Kr_2^+ + SF_6^- + M \rightarrow KrF^* + \text{other products}$$

The subsequent decrease in α at higher gas pressures is believed to be due to the ionic recombination reaction becoming diffusion controlled.

Comparison with Theory. Bates and co-workers[11] have determined the recombination rate constant pressure profile for the reaction

$$O_4^+ + O_4^- + O_2 \xrightarrow{\alpha_s} [O_8] + O_2$$

over the pressure region applicable to the Kr_2^+/SF_6^- system. Having calculated the pressure dependence for this reaction, they showed that scaling parameters, λ and η, existed so that rate constants for any ionic recombination reaction, X, could be obtained from the standard parameters, S, by

$$\lambda\alpha_X = \alpha_S \qquad \eta N_X = N_S$$

where the scaling parameters are given by

$$\lambda' = \left(\frac{A_S B_S}{A_X B_X}\right)^{1/2} \qquad \eta = \left(\frac{A_X B_S}{A_S B_X}\right)^{1/2}$$

with

$$\lambda = \lambda'(1 + (\Omega^{1/2})/6)$$

where

$$\Omega = \left|\frac{M_1 - M_2}{M_1 + M_2}\right|$$

$$A_S = 2.54 \times 10^{-6}\ \text{cm}^3\ \text{s}^{-1}$$

$$B_S = 4.73\ \text{cm}^2\ \text{V}^{-1}\ \text{s}^{-1}$$

$$A_X = 1.087 \times 10^{-5} P^{1/2}\left(\frac{300}{T}\right)^3\left(\frac{F(a_1)}{M_{13}{}^{1/2}} + \frac{F(a_2)}{M_{23}{}^{1/2}}\right)$$

$$B_X = K_1 + K_2 = 13.82 P^{-1/2}(M_{13}{}^{-1/2} + M_{23}{}^{-1/2})$$

and M_1 is the mass of the positive ion, M_2 the mass of the negative ion, M_3 the mass of the neutral third body, P the polarizability of the bulk medium, M_{13} the positive ion–third body reduced mass, M_{23} the negative ion–third body reduced mass, K_i the mobility of ion i, and $F(a_i)$ the interaction potential values, calculated assuming a polarization interaction[11] with

$$a_i = \frac{M_1 + M_2 - M_i}{M_i(M_1 + M_2 + M_3)}$$

For the reaction

$$Kr_2^+ + SF_6^- + Kr \rightarrow KrF^* + \text{products}$$

the parameters used were

$$M_1 = \text{mass of } Kr_2^+ = 167.6\ \text{amu}$$

$$M_2 = \text{mass of } SF_6^- = 146.1\ \text{amu}$$

$$M_3 = \text{mass of Kr} = 83.8\ \text{amu}$$

The positive ion was taken as the dimer, for the experimental reasons explained earlier.

The B_X parameter, being the sum of the two ion mobilities, can be replaced by experimentally determined values. The Kr_2^+ mobility in krypton has been measured[28] as 0.995 $\text{cm}^2\ \text{V}^{-1}\ \text{s}^{-1}$. To account for the lower mobilities of the monomer ions, e.g., 0.806 (Kr^+, $^3P_2$) and 0.866 (Kr^+, $^1P_2$) $\text{cm}^2\ \text{V}^{-1}\ \text{s}^{-1}$ [29] an "averaged" value of 0.90 was used for the cation mobility.

For the negative ion, SF_6^-, no experimental determination of the mobility could be found. From the literature[30] the mobility for this ion in krypton can be calculated as $\mu^- \approx 1.20\ \text{cm}^2\ \text{V}^{-1}\ \text{s}^{-1}$.

For the Bates standard oxygen reaction, the calculated ionic mobilities were found to be too high in comparison with experimental data and computer simulation predictions.[11] To compensate for this, a value of 1.0 $\text{cm}^2\ \text{V}^{-1}\ \text{s}^{-1}$ was taken as the mobility of SF_6^- in krypton.

Thus using the values of

$$A_X = 2.157 \times 10^{-6}\ \text{cm}^3\ \text{s}^{-1}$$

$$B_X = 1.90\ \text{cm}^2\ \text{V}^{-1}\ \text{s}^{-1}$$

gives as final parameters

$$\lambda = 1.79 \qquad \eta = 1.45$$

These values were used to calculate rate constants over the experimental pressure range, and the results obtained, α_{BATES}, are compared to the experimental data in Figure 4a. As found for the Xe_2^+/SF_6^- gas system,[8] the experimental data are seen to be much greater in magnitude than the termolecular theory predictions and have an experimental maximum at a lower pressure than predicted.

As the agreement between the experimental and termolecular values was so poor, values for the reaction

$$Kr_2^+ + F^- + Kr \rightarrow KrF^* + 2Kr$$

were also calculated. The F^- ion has been shown to result from the fragmentation of SF_6^-,[31] although on a much longer time scale than used here. The experimentally determined F^- mobility in krypton is $\mu^- = 2.30\ \text{cm}^2\ \text{V}^{-1}\ \text{s}^{-1}$,[32] which gives

$$\lambda = 1.31 \qquad \mu = 1.30$$

These values are plotted against the experimental data in Figure 4b. Although the general agreement in the magnitude of the rate constant is better, the peak value is still predicted at too high a pressure, and the calculated curve crosses the experimental data at the high pressure end.

Modifications to the Termolecular Theory. Several modifications to the Bates termolecular theory that were explored in the previous investigation[8] have also been considered here. It was found that the effect of the krypton dimer ion taking part in symmetrical resonance charge transfer[30] according to the reaction

$$Kr\text{–}Kr^+ + Kr \rightarrow Kr + Kr^+\text{–}Kr$$

and the contribution from the ionic recombination reaction with an ion as the third body[33] (for a maximum ion concentration of $\sim 10^{-8}$ M)

$$Kr_2^+ + SF_6^- + Kr_2^+ \text{ (or } SF_6^-) \rightarrow KrF^* + \text{products}$$

were negligible.

To explain the large discrepancy between experiment and theory at low and medium pressures for Kr_2^+/SF_6^- recombination, *two-body*, mutual neutralization reactions are proposed to be occurring. By analogy to the Xe_2^+/SF_6^- recombination system,[8] where up to 60% of the total ionic recombination proceeded by this pathway, this reaction would correspond to cation neutralization by a F^- ion transfer of the form

$$Kr_2^+ \cdots {}^-F\text{–}SF_5 \rightarrow KrF^* + \text{products}$$

This process, which may be regarded as occurring through an avoided crossing between an ionic and covalent potential surface,[11] has been shown to be considerably enhanced by the presence of

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a bulk gas, which can collisionally convert free ions to bound ion pairs, thus allowing the avoided crossing to be traversed many times.

It was unfortunately beyond the scope of this study to calculate the effect of mutual neutralization reactions in Kr_2^+/SF_6^- recombination. Qualitative estimates of the fraction of reaction in this pathway were attempted by matching the pressure profile experimentally observed in this study to the theoretical curves given for the standard Bates O_4^+/O_4^- calculations.[11] This gave a very large prediction of ~80% of the recombination proceeding by this pathway.

(b) Xe_2^+/Cl^- Ionic Recombination. To contrast the Kr_2^+/SF_6^- study, where the recombination involved a molecular anion, the Xe_2^+/Cl^- system was also studied. The XeCl* exciplex was produced by the pulsed electron irradiation of xenon/$CFCl_3$ gas mixtures. $CFCl_3$ is a good electron thermalizer and is known to dissociatively capture thermal electrons to produce Cl^-.[34,35] However, its ionization potential (11.77 eV[36]) is lower than that of xenon (12.130 eV[24]); thus the charge-transfer reaction

$$Xe^+ + CFCl_3 \rightarrow Xe + (CFCl_3)^+$$

occurs in competition with the recombination reaction

$$Xe_2^+ + Cl^- + Xe \rightarrow XeCl^* + 2Xe$$

This experimental problem was minimized by using a very small pressure of the halide source gas, typically 0.10 Torr, and keeping the ion concentrations as high as possible, usually greater than 5×10^{-8} M. This maximized the rate of the ion recombination reaction, relative to the charge-transfer reaction which has a $\tau_{1/2} > 1$ μs under these experimental conditions.

The XeCl* emission spectrum for the $^2\Sigma^+_{1/2} \rightarrow {}^2\Sigma^+_{1/2}$ transition covers the range 294–314 nm[12] and has a maximum at 308 nm. All kinetic measurements were done at the peak wavelength.

The emission curves again contained components from X-rays, background dimer, and XeCl* fluorescence from neutral reaction and ionic combination. The X-ray and dimer rare-gas fluorescence was subtracted as described for Kr_2^+/SF_6^-, with the background wavelength used being 316 nm.

In this system, however, the relative contributions of the rare-gas neutral excited-state reaction and ionic recombination processes could not be resolved. The direct reaction fluorescence could be isolated by lowering the incident pulse energy to extremely low values and by using very low xenon pressures. This slowed the ionic recombination reaction to such an extent that at short times (<200 ns) the fluorescence from this process was negligible. However, integration of these kinetic curves to obtain yields of these two processes proved to be irreproducible.

Thus the excited state yields for XeCl* were *calculated* by measurement of the isolated excited state formed fluorescence rate constants as a function of xenon and $CFCl_3$ pressures and assuming that the reaction scheme presented for the Kr* formation of KrF* would also be applicable in the XeCl* system.

This gave the following values:

$$Xe^* + CFCl_3 \rightarrow XeCl^* \qquad k_h = 1.40 \times 10^{11}\ M^{-1}\ s^{-1} \qquad (h)$$

$$Xe^* + Xe \rightarrow \text{products} \qquad k_i = 5.47 \times 10^8\ M^{-1}\ s^{-1} \qquad (i)$$

$$Xe^* \rightarrow h\nu \qquad k_j = 2.10 \times 10^6\ s^{-1} \qquad (j)$$

$$XeCl^* + CFCl_3 \rightarrow \text{products} \qquad k_k = 2.77 \times 10^{11}\ M^{-1}\ s^{-1}\ [37,38] \qquad (k)$$

$$XeCl^* + Xe \rightarrow \text{products} \qquad k_l = 6.02 \times 10^9\ M^{-1}\ s^{-1}\ [39] \qquad (l)$$

$$XeCl^* + 2Xe \rightarrow \text{products} \qquad k_m = 2.65 \times 10^{11}\ M^{-2}\ s^{-1}\ [38] \qquad (m)$$

$$XeCl^* \rightarrow h\nu \qquad k_n = 9.09 \times 10^7\ s^{-1}\ [40] \qquad (n)$$

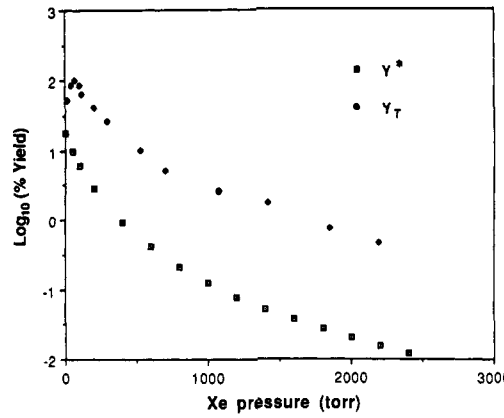


Figure 5. Xenon pressure dependence of the measured total (Y_T) and calculated excited-state reaction (Y^*) yields for Xe_2^+/Cl^- recombination.

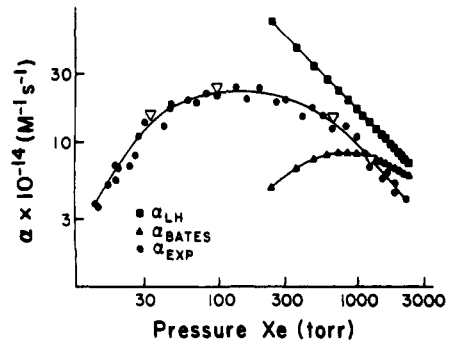


Figure 6. Comparison of the experimental rate constants, α_{EXP}, to the termolecular Bates, α_{BATES}, and diffusion-controlled Langevin–Harper, α_{LH}, calculations for the reaction $Xe_2^+ + Cl^- + Xe \xrightarrow{\alpha}$ products. Also shown are the ionic recombination rate constants obtained from irradiated Xe/CF_2Cl_2 gas mixtures (▽).

The obtained Y^* values are shown in Figure 5, in contrast to the experimentally determined total yield (Y_T) values over the pressure range of this study. The overall subtractions for this system are seen to be small, the largest being ~30% at a xenon pressure of 20 Torr and much less at higher pressures. This is in marked contrast to the KrF* system, where corrections up to 70% of the integrated yield were made at low pressures.

The rate coefficients, α, as a function of xenon pressure, are shown in Figure 6. The experimental curve has the same shape as for KrF*, but exhibits a much broader peak, stretching over several hundred Torr. The magnitude of the rate constants are comparable to the values for the KrF* exciplex, but the maximum for this exciplex is at a lower gas pressure.

Comparison with Theory. The calculation of the standard termolecular recombination rate constants using the Bates formulae[11] for the reaction

$$Xe_2^+ + Cl^- + Xe \rightarrow XeCl^* + 2Xe$$

give

$$A_X = 2.97 \times 10^{-6}\ cm^3\ s^{-1} \qquad B_X = 2.04\ cm^2\ V^{-1}\ s^{-1}$$

Using the determined mobilities, 0.60 $cm^2\ V^{-1}\ s^{-1}$ for Xe_2^+[8] and 1.34 $cm^2\ V^{-1}\ s^{-1}$[41] for Cl^- in Xe, gives as final parameters

$$\lambda = 1.65 \qquad \eta = 1.69$$

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The termolecular values (α_{BATES}), along with the Langevin–Harper, (α_{LH}), values at the same pressures are contrasted to the experimental data in Figure 6.

As for Kr_2^+/SF_6^-, the Xe_2^+/Cl^- experimental rate constants are again greater in magnitude than the values calculated from the termolecular theory and peak at a much lower xenon pressure than predicted. The termolecular theory modifications of symmetrical resonance charge transfer and ion third body recombination were again found to be negligible.

The theoretical values for the reaction

$$Xe_2^+ + CFCl_3^- + Xe \rightarrow XeCl^* + \text{products}$$

were not calculated, as this ion is only formed with an extremely low efficiency, ~0.2%.[42] Other molecular anions are also formed,[43] but only by the capture of higher energy electrons, e.g., Cl_2^- at electron energies of 1.6 and 3.0 eV and CCl_3^- at 3.3 eV. However, these ions also have very low yields. The major product from the $CFCl_3$ capture of *thermal* electrons is Cl^-.

It can also be shown that the recombining anion must be formed directly upon electron capture. Complexes such as $Cl{\cdot}CFCl_3$ cannot be formed on the experimental time scales and then recombine with the rare-gas cations. Under typical experimental conditions, consisting of a $CFCl_3$ pressure of 0.10 Torr (5.41×10^{-6} M), and an electron concentration of $\sim 5 \times 10^{-8}$ M, assuming a maximum collisional reaction rate constant for complex formation of 3×10^{11} M^{-1} s^{-1} gives a half-life for formation as ~430 ns; which is much longer than the observed exciplex fluorescence.

The discrepancy between experiment and theory for Xe_2^+/Cl^- recombination cannot be the ion abstraction proposed for Kr_2^+/SF_6^-. Two-body reactions are possible, such as electron abstraction (transfer), according to

$$Xe_2^+ + Cl^- \rightarrow Xe_2^* + Cl$$

but this reaction does not produce the fluorescent XeCl* exciplex. A consecutive reaction mechanism for XeCl* production such as

$$Xe_2^+ + Cl^- \rightarrow Xe_2^* + Cl$$

$$Xe_2^* + CFCl_3 \rightarrow XeCl^* + \text{products}$$

can also be eliminated by using the same arguments that showed that the complex anion $^-Cl{\cdot}CFCl_3$ could not be involved in the reaction. Furthermore, the broad pressure profile observed in this study is sufficiently different from the Bates standard mutual neutralization curves[11] that no quantitative or qualitative comparison is possible. This system is under theoretical investigation[44] and will be reported shortly.

In summary, the three-body ionic recombination mechanism predictions do not model the experimental pressure profiles accurately. While it is possible to explain the large discrepancy for the Kr_2^+/SF_6^- system, by invoking a significant proportion of two-body, mutual neutralization, recombination, no explanation can currently be given for the different behavior of the Xe_2^+/Cl^- system.

(2) *High Pressures.* At higher gas pressures, the Langevin–Harper[9,10] diffusion formula

$$\alpha_{LH} = 4\pi e(\mu^+ + \mu^-)$$

holds. The sum $(\mu^+ + \mu^-)$ is equivalent to the B_X parameter of the termolecular recombination calculation previously, and hence for KrF*

$$\alpha_{LH} = 1.8 \times 10^{-6} B_X \text{ cm}^3 \text{ s}^{-1}$$

In comparison with the Langevin–Harper calculations, for SF_6^- as the recombining anion (Figure 4a, α_{LH}), excellent agreement is found with the experimental data at very high pressures (>1000 Torr). Within error, the experimental data converge to the calculated limit, showing that under these conditions the two ions *always* recombine to give the fluorescent exciplex once they have diffused together.

The Langevin–Harper values for F^- as the recombining anion (Figure 4b) are sufficiently close to the values for the Kr_2^+/SF_6^- system that no definitive conclusion on the identity of the anion can be made from this calculation.

The Xe_2^+/Cl^- recombination shows a limiting high-pressure slope that is parallel to, but *lower* than, the Langevin–Harper values. Typically the diffusion-controlled experimental rate constants are only about 50% of the theoretical predictions. This discrepancy is much larger than the experimental error associated with these measurements (<20%). Such behavior implies that XeCl* is being formed by a diffusion-controlled reaction at these pressures but that half the ions are *not* recombining upon the bound system being formed. This is attributed to collisions between neutral atoms and one of the ions in the bound state, which can raise the system energy above zero to again give free ions.

Other Experiments. To further identify the recombining anion for XeCl*, experiments were also performed using pulsed electron irradiated gas mixtures of xenon and CF_2Cl_2. This molecule is also known to dissociatively capture thermal electrons to produce Cl^-,[34,35] and its ionization potential (12.31 eV[35]) is higher than that of xenon, eliminating the interfering charge transfer reaction that occurs for $CFCl_3$.

The pressure of CF_2Cl_2 used was kept constant at 0.10 Torr, and the xenon pressure varied over the pressure range 50–1000 Torr. The only corrections made for this system were for X-ray and background rare-gas dimer fluorescence as the neutral reaction formed exciplex was negligible at this halide source pressure.

The results obtained are also shown in Figure 6 (▽) in contrast to the $CFCl_3$ values. As can be seen, excellent agreement was obtained between the two experimental sets of data, further confirming that Cl^-, produced by the thermal electron capture of $CFCl_3$, is the recombining anion.

Conclusion

The experimental rate constants at low/medium rare-gas pressures for Kr_2^+/SF_6^- and Xe_2^+/Cl^- recombination are found not to be accurately modeled by the Bates[11] termolecular recombination theory. The predicted rate constants are far lower than observed experimentally, and the calculated maxima are at much higher gas pressures. The previously attempted extensions of this theory fail to give any significant improvement in agreement when applied to these systems.

For Kr_2^+/SF_6^- recombination, this discrepancy is attributed to a two-body charge neutralization process, where an F^- ion is abstracted from the complex anion by the rare-gas dimer cation, to produce the fluorescent KrF* exciplex. However, this mechanism cannot be invoked for Xe_2^+/Cl^- recombination, as it would not produce the observed fluorescent exciplex.

The agreement between experiment and theory at high pressures for Kr_2^+/SF_6^- recombination is excellent, with the exciplex being formed at a diffusion-controlled rate and always recombining to give the exciplex upon the ions diffusing together. In contrast, the Xe_2^+/Cl^- recombination at high pressures is seen to be diffusion-controlled but to recombine with only approximately 50% efficiency.

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Registry No. Kr_2^+, 37240-74-7; SF_6^-, 25031-39-4; Xe_2^+, 12185-20-5; Cl^-, 16887-00-6.

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